

dgnozzle

A discontinuous Galerkin solver for planar nozzle design

Formulation, geometry and verification

Abstract

This document defines the problem **dgnozzle** solves and the scheme it uses. The governing equations are the two-dimensional compressible Euler equations, discretised by a nodal discontinuous Galerkin method on triangles or quadrilaterals with a Roe numerical flux, and marched to steady state in pseudo-time. The geometry is a planar converging-diverging nozzle whose wall is parameterised by a small number of design variables. The solver is intended for *design studies*: the reader is expected to change the geometry, sweep it, and take sensitivities, not to modify the discretisation. Section 11 reports the verification evidence, including observed orders of accuracy.

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1 Scope and conventions

The solver computes steady, inviscid, compressible flow through a planar converging–diverging nozzle. “Planar” means the channel is two-dimensional and of unit depth, not axisymmetric. One consequence matters throughout: the one-dimensional *area* is the channel height, so

$$A(x) = 2 y_{\text{wall}}(x), \quad \frac{A_{\text{exit}}}{A_{\text{throat}}} = \frac{y_{\text{wall}}(L)}{y_{\text{throat}}} \equiv \text{AR}. \quad (1)$$

Isentropic tables may be used directly, because the area–Mach relation depends on A/A^* and not on the cross-section’s shape.

Only the upper half of the channel is meshed; the lower half follows by symmetry about $y = 0$. All reported *integral* quantities (thrust, mass flow) are doubled to describe the full channel per unit depth.

Non-dimensionalisation. The reference state is the inlet reservoir. With $p_t = 1$, $T_t = 1$, $R = 0.4$ and $\gamma = 1.4$ the stagnation speed of sound and density are

$$a_t = \sqrt{\gamma R T_t} = 0.74833, \quad \rho_t = \frac{\gamma p_t}{a_t^2} = 2.5. \quad (2)$$

These are the units of the original MATLAB solver, so results are directly comparable.

2 Governing equations

The two-dimensional Euler equations in conservative form are

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = \mathbf{0}, \quad (3)$$

with

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ \rho u H \end{pmatrix}, \quad \mathbf{G} = \begin{pmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ \rho v H \end{pmatrix}. \quad (4)$$

The system is closed by the ideal-gas relation and the definition of total enthalpy,

$$p = (\gamma - 1) \left(\rho E - \frac{1}{2} \rho (u^2 + v^2) \right), \quad H = E + \frac{p}{\rho}, \quad a = \sqrt{\frac{\gamma p}{\rho}}. \quad (5)$$

It is convenient to write the flux projected onto a direction $\mathbf{n} = (n_x, n_y)$,

$$\mathbf{F} \cdot \mathbf{n} \equiv \mathbf{F}n_x + \mathbf{G}n_y = \begin{pmatrix} \rho u_n \\ \rho u u_n + p n_x \\ \rho v u_n + p n_y \\ \rho H u_n \end{pmatrix}, \quad u_n = u n_x + v n_y, \quad (6)$$

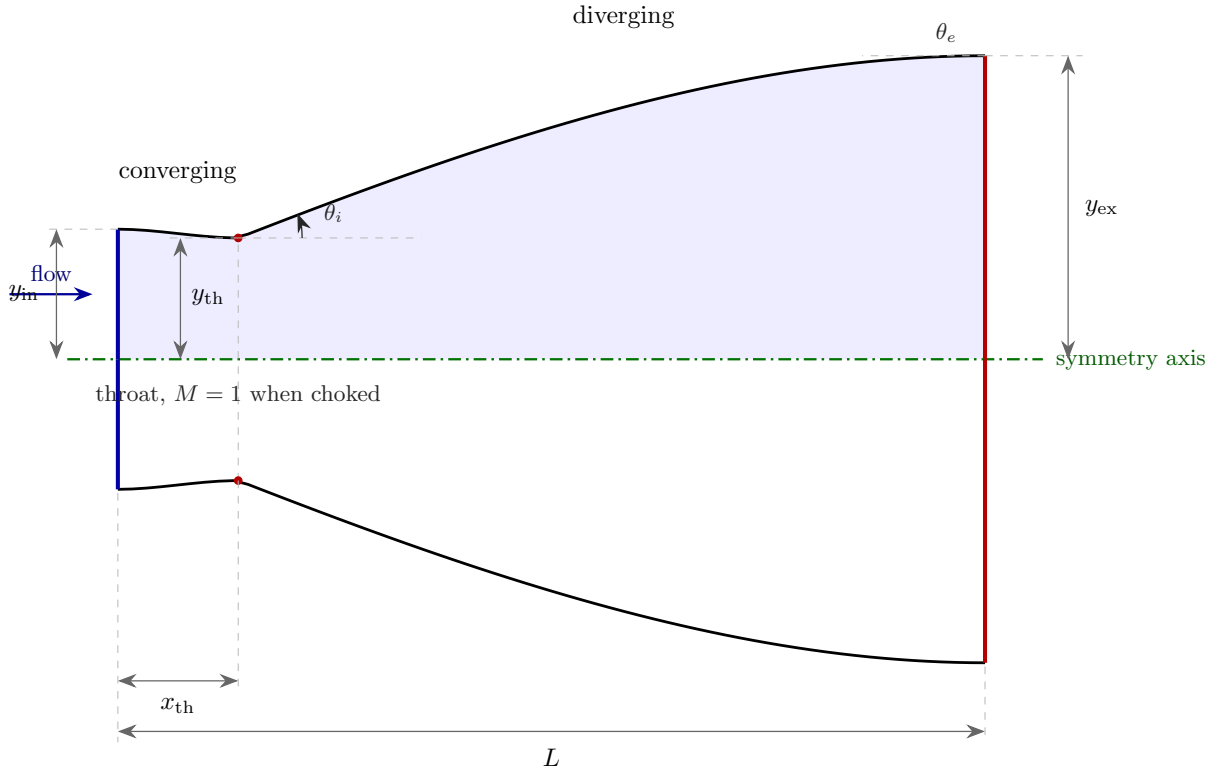
since every boundary and interface term below is of this form.

The flux Jacobian $\partial(\mathbf{F} \cdot \mathbf{n})/\partial \mathbf{U}$ has eigenvalues

$$\lambda = \{ u_n + a, u_n - a, u_n, u_n \}, \quad (7)$$

whose signs determine how many conditions may be imposed at a boundary (Section 6).

3 Nozzle geometry



Boundary conditions

- INFLOW — subsonic stagnation: T_t, p_t , flow angle α
- WALL — inviscid slip, $\vec{v} \cdot \vec{n} = 0$ (thrust integrated here)
- .-.- AXIS — symmetry; same flux as WALL, tagged apart
- OUTFLOW — static p_b if subsonic, extrapolation if supersonic
- meshed half-domain (the lower half follows by symmetry)

Figure 1: Nozzle geometry, design variables and boundary conditions. The shaded region is the half-domain that is actually meshed. The wall shown is the ‘bell’ contour at $AR = 2.5019$, $x_{th} = 0.1388$.

3.1 Design variables

The wall half-height $y_{wall}(x; \mathbf{a})$ is controlled by

$$\mathbf{a} = (AR, x_{th}, y_{th}, y_{in}, L, \theta_i, \theta_e, w_1, w_2), \quad (8)$$

listed in Table 1. Every one of them is differentiable (Section 12).

Symbol	Code name	Meaning
AR	area_ratio	exit-to-throat area ratio, Eq. (1)
x_{th}	throat_x	throat location as a fraction of L
y_{th}	throat_half_height	throat half-height [m]; scales the nozzle
y_{in}	inlet_half_height	inlet half-height [m]; must exceed y_{th}
L	length	axial length [m]
θ_i	theta_initial_deg	wall angle just downstream of the throat
θ_e	theta_exit_deg	wall angle at the exit plane
w_1, w_2	bezier_w1/w2	Bézier shape weights

Table 1: Design variables.

3.2 Converging section

For $x \leq x_{\text{th}}L$, with $s = x/(x_{\text{th}}L) \in [0, 1]$, the wall is a cubic Hermite with zero slope at both ends,

$$y_{\text{wall}}(s) = h_{00}(s)y_{\text{in}} + h_{01}(s)y_{\text{th}}, \quad \begin{aligned} h_{00}(s) &= 2s^3 - 3s^2 + 1, \\ h_{01}(s) &= -2s^3 + 3s^2. \end{aligned} \quad (9)$$

Because $y'(x_{\text{th}}^-) = 0$, the point x_{th} really is a stationary point of the wall, so “throat” and “ x_{th} ” agree.

3.3 Diverging section

For $x > x_{\text{th}}L$, with $s = (x - x_{\text{th}}L)/(L - x_{\text{th}}L)$ and $y_{\text{ex}} = \text{AR} y_{\text{th}}$, five families are available.

‘conical’ — straight wall.

$$y_{\text{wall}} = y_{\text{th}} + (y_{\text{ex}} - y_{\text{th}})s. \quad (10)$$

‘bell’ and ‘moc’ — prescribed wall angles. A cubic Hermite with slopes $\tan \theta_i$ and $\tan \theta_e$ (converted to the s variable through dx/ds):

$$y_{\text{wall}} = h_{00}y_{\text{th}} + h_{10}m_i + h_{01}y_{\text{ex}} + h_{11}m_e, \quad \begin{aligned} h_{10}(s) &= s^3 - 2s^2 + s, \\ h_{11}(s) &= s^3 - s^2, \end{aligned} \quad (11)$$

with $m_i = \tan \theta_i \cdot dx/ds$ and similarly for m_e . ‘moc’ fixes $\theta_i = 30^\circ$, $\theta_e = 0$, the classical Rao values.

A slope discontinuity, on purpose. With $\theta_i > 0$ the wall slope jumps from 0 to $\tan \theta_i$ at the throat. That sharp corner is the idealisation used in minimum-length nozzle design — it launches a centred Prandtl–Meyer expansion fan — and it is *not* a defect. It does, however, put a singularity into the exact solution, which caps the achievable order of accuracy. Section 11 quantifies this. Use ‘smooth’ or ‘analytic’ for convergence studies.

‘smooth’ — C^1 throat. Equation (11) with $\theta_i = 0$, so the wall slope is continuous across the throat and the exact solution is smooth.

'bezier' — two shape weights. A cubic Bézier in s with interior control values $c_k = y_{\text{th}} + w_k(y_{\text{ex}} - y_{\text{th}})$:

$$y_{\text{wall}} = (1-s)^3 y_{\text{th}} + 3(1-s)^2 s c_1 + 3(1-s) s^2 c_2 + s^3 y_{\text{ex}}. \quad (12)$$

$(w_1, w_2) = (\frac{1}{3}, \frac{2}{3})$ reproduces the conical wall; $w_1 = 0$ gives a C^1 throat. This is the most convenient family for gradient-based shape optimisation, because w_1 and w_2 are bounded, dimensionless, and act locally.

'analytic' — the legacy verification contour. The fixed shape of the original MATLAB solver,

$$t(x) = \pi \ln(1 + (e^2 - 1)x/L), \quad y_{\text{wall}} = 0.01 \left(4 - \cos \frac{t}{2}\right) \left(\cos t + 5 - \cos \frac{t}{2}\right), \quad (13)$$

which runs from $y = 0.15$ at the inlet through a throat of 0.13989434 at $x = 0.138805$ to $y = 0.35$ at the exit, an area ratio of 2.50189. It ignores `area_ratio` and `throat_x`; it exists so that results can be checked against the original code.

4 Discretisation

4.1 Mesh

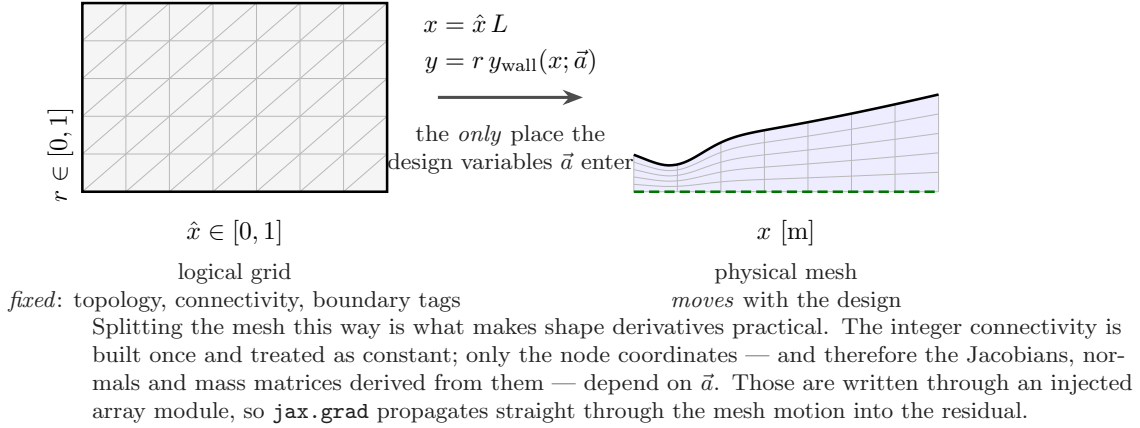


Figure 2: The mesh is built as a fixed logical grid plus a design-dependent map. Only the map depends on $\mathbf{a}$, which is what makes shape derivatives tractable.

A structured $n_x \times n_r$ grid of cells is laid out in logical coordinates $(\hat{x}, r) \in [0, 1]^2$ and mapped to the nozzle by

$$x = \hat{x} L, \quad y = r y_{\text{wall}}(x; \mathbf{a}). \quad (14)$$

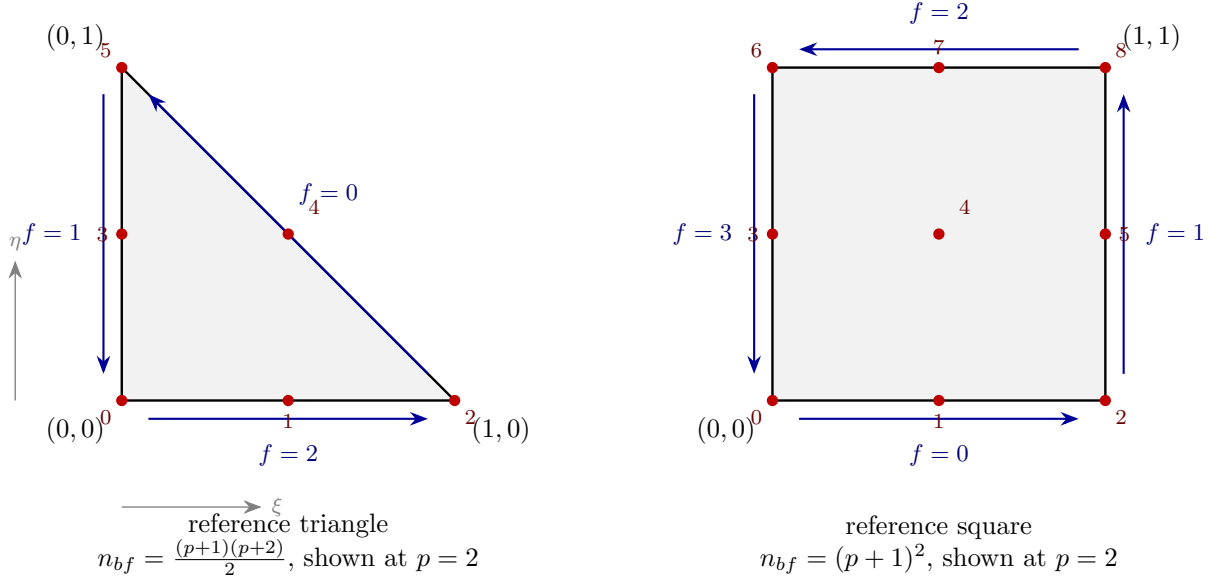
Each cell becomes one quadrilateral or two triangles. Refinement level ref subdivides each cell 2^{ref} ways per direction, so the element count is $n_x n_r 4^{\text{ref}}$ (doubled for triangles).

Axial nodes may be clustered toward the throat, where the solution varies fastest. The base distribution is generated from a target spacing

$$h(\hat{x}) = 1 + (\sigma - 1) \left(1 - e^{-((\hat{x} - x_{\text{th}})/w)^2}\right), \quad (15)$$

integrated and inverted, with σ the far-field-to-throat spacing ratio. Refinement then subdivides that base distribution linearly, so the node set at one refinement level is a subset of the next.

4.2 Reference elements



Vertices run counter-clockwise, so the tangent of a local edge, rotated by -90° , is the *outward* normal. Nodes are stored with η (or j) outermost: **for iy: for ix**. The neighbour across an interior edge traverses it backwards, so its quadrature parameter is $1 - \sigma$ — which is what makes quadrature point q of both sides land on the same physical location.

Figure 3: Reference elements, Lagrange node ordering and local edge numbering.

The solution is expanded in a nodal Lagrange basis of order p on each element,

$$\mathbf{U}_h(\mathbf{x})|_{\Omega_e} = \sum_{i=1}^{n_{bf}} \mathbf{U}_{e,i} \phi_i(\xi, \eta), \quad n_{bf} = \begin{cases} \frac{1}{2}(p+1)(p+2) & \text{triangle,} \\ (p+1)^2 & \text{quadrilateral.} \end{cases} \quad (16)$$

The element geometry uses an independent Lagrange basis of order Q , so $Q = 1$ gives straight-sided elements and $Q = 2$ a quadratic (curved) wall.

The mapping from reference to physical coordinates and its Jacobian are

$$\mathbf{x}(\xi, \eta) = \sum_n \mathbf{x}_n \psi_n(\xi, \eta), \quad J = \frac{\partial(x, y)}{\partial(\xi, \eta)}, \quad \nabla_{\mathbf{x}} \phi_i = J^{-T} \nabla_{(\xi, \eta)} \phi_i. \quad (17)$$

4.3 Weak form

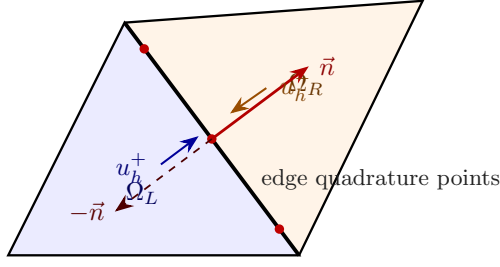
Multiplying Eq. (3) by a test function ϕ_i , integrating over an element and applying the divergence theorem gives the discrete statement

$$\boxed{\int_{\Omega_e} \phi_i \frac{\partial \mathbf{U}_h}{\partial t} d\Omega = \int_{\Omega_e} \nabla \phi_i \cdot \mathcal{F}(\mathbf{U}_h) d\Omega - \oint_{\partial\Omega_e} \phi_i \hat{F}(\mathbf{U}_h^+, \mathbf{U}_h^-, \mathbf{n}) ds} \quad (18)$$

for every basis function i and every element e . Writing the right-hand side as $-\mathbf{R}_e$ leaves the semi-discrete system

$$M_e \dot{\mathbf{U}}_e = -\mathbf{R}_e(\mathbf{U}), \quad (M_e)_{ij} = \int_{\Omega_e} \phi_i \phi_j d\Omega. \quad (19)$$

The mass matrix is block diagonal — an element couples to itself only — so M^{-1} is applied as n_{elem} independent $n_{bf} \times n_{bf}$ solves.



At each edge quadrature point the two elements present *different* values, u_h^+ and u_h^- . A Roe solve on that one-dimensional Riemann problem returns a single numerical flux

Ω_L owns the edge and adds $\oint \hat{F}(u_h^+, u_h^-, \vec{n}) \cdot \vec{n} \phi_i \hat{F} ds$; Ω_R sees $-\vec{n}$ and adds $-\oint \phi_i \hat{F} ds$. The same number enters both with opposite sign, so whatever leaves one element enters the other exactly — which is why the scheme is conservative.

In the code the owner is the element with the lower index, and every `(element, local face)` pair maps to one global edge through `face_edge`, with `face_sign` carrying the $\pm$. Assembly is then a pure *gather*: no element ever writes into another's storage, so the loop parallelises without atomics and runs unchanged under NumPy, Numba and JAX.

Figure 4: Elements communicate only through the numerical flux on their shared edge. The same value enters both with opposite sign, which is what makes the scheme conservative.

4.4 Quadrature

Volume integrals use symmetric positive-weight rules on the triangle (a Duffy-collapsed Gauss–Jacobi rule for degree ≥ 3) and tensor-product Gauss–Legendre on the quadrilateral; edge integrals use Gauss–Legendre on $[0, 1]$. The degrees are

$$d_{\text{vol}} = 2p + 1 + 2(Q - 1), \quad d_{\text{edge}} = 2p + Q. \quad (20)$$

Positive weights are not optional. The Dunavant degree-3 rule used by the original code has a negative first weight, -0.28125 . A negative weight can destroy the positive definiteness of M_e and lets the entropy-error integral — a sum of squares — go negative. Every rule here has strictly positive weights, checked against exact monomial integrals in the test suite.

5 Numerical flux

The interface flux is Roe's approximate Riemann solver,

$$\hat{F}(\mathbf{U}^+, \mathbf{U}^-, \mathbf{n}) = \frac{1}{2} (\mathbf{F}(\mathbf{U}^+) \cdot \mathbf{n} + \mathbf{F}(\mathbf{U}^-) \cdot \mathbf{n}) - \frac{1}{2} |\hat{A}| (\mathbf{U}^- - \mathbf{U}^+), \quad (21)$$

where $\hat{A}$ is the flux Jacobian evaluated at the Roe-averaged state

$$\hat{u} = \frac{\sqrt{\rho^+} u^+ + \sqrt{\rho^-} u^-}{\sqrt{\rho^+} + \sqrt{\rho^-}}, \quad (\text{and likewise for } \hat{v}, \hat{H}), \quad \hat{a}^2 = (\gamma - 1) \left(\hat{H} - \frac{1}{2} \hat{q}^2 \right). \quad (22)$$

The dissipation term is evaluated in factored form rather than by assembling $|\hat{A}|$: with $\Delta(\cdot) = (\cdot)^- - (\cdot)^+$,

$$G_1 = (\gamma - 1) \left(\frac{1}{2} \hat{q}^2 \Delta \rho - \hat{u} \Delta(\rho u) - \hat{v} \Delta(\rho v) + \Delta(\rho E) \right), \quad (23)$$

$$G_2 = -\hat{u}_n \Delta \rho + \Delta(\rho u) n_x + \Delta(\rho v) n_y, \quad (24)$$

$$s_1 = \frac{1}{2} (|\lambda_1| + |\lambda_2|), \quad s_2 = \frac{1}{2} (|\lambda_1| - |\lambda_2|), \quad (25)$$

$$C_1 = \frac{G_1}{\hat{a}^2} (s_1 - |\lambda_3|) + \frac{G_2}{\hat{a}} s_2, \quad C_2 = \frac{G_1}{\hat{a}} s_2 + (s_1 - |\lambda_3|) G_2. \quad (26)$$

Entropy fix. An unmodified Roe flux admits a non-physical expansion shock where an eigenvalue passes through zero — which happens at the throat of every choked nozzle. The Harten–Hyman fix replaces small eigenvalues by a smooth quadratic,

$$|\lambda| \leftarrow \begin{cases} \frac{\lambda^2 + \epsilon^2}{2\epsilon}, & |\lambda| < \epsilon, \\ |\lambda|, & \text{otherwise,} \end{cases} \quad \epsilon = 0.05 \hat{a}. \quad (27)$$

6 Boundary conditions

The number of conditions that may be imposed equals the number of *incoming* characteristics, read off the eigenvalues in Eq. (7) with $\mathbf{n}$ the outward normal.

6.1 Inviscid wall and symmetry axis

Both are impermeable: $\mathbf{v} \cdot \mathbf{n} = 0$. Removing the normal velocity and recovering the pressure from the remaining energy gives a flux that transmits pressure but no mass or energy,

$$p_b = (\gamma - 1) \left(\rho E - \frac{1}{2} \rho |\mathbf{v}_{\parallel}|^2 \right), \quad \hat{F} = (0, p_b n_x, p_b n_y, 0)^T. \quad (28)$$

The contoured wall and the symmetry axis use the same flux but carry different tags, so that thrust can be integrated over the wall alone.

6.2 Subsonic stagnation inflow

Three characteristics enter, one leaves. The outgoing one carries the interior Riemann invariant

$$J^+ = u_n + \frac{2a}{\gamma - 1}. \quad (29)$$

Combining it with the isentropic stagnation relations and a prescribed flow direction $\hat{d} = (\cos \alpha, \sin \alpha)$ gives a quadratic for the inflow Mach number,

$$\underbrace{\left(\frac{\gamma-1}{2} \beta - n_d^2 \right)}_A M^2 - \underbrace{\frac{4n_d}{\gamma-1}}_B M + \underbrace{\left(\beta - \left(\frac{2}{\gamma-1} \right)^2 \right)}_C = 0, \quad \beta = \left(\frac{J^+}{a_t} \right)^2, \quad n_d = \mathbf{n} \cdot \hat{d}. \quad (30)$$

The *smallest non-negative* root is the physical branch; the boundary state then follows from the isentropic relations and total enthalpy $H_b = a_t^2/(\gamma - 1)$.

Root selection. The legacy implementation always took $(-B + \sqrt{B^2 - 4AC})/2A$. That is the physical root only while $A > 0$; when the leading coefficient changes sign — which happens for weak inflow — it returns a negative Mach number and the boundary state becomes meaningless.

6.3 Outflow

If $u_n/a \geq 1$ the outflow is supersonic: no characteristic enters, nothing may be imposed, and the interior state is extrapolated. If it is subsonic, exactly one condition is admissible. Static pressure is set to p_b , while entropy and the outgoing invariant are carried from the interior and the tangential velocity is unchanged:

$$\rho_b = \rho^+ \left(\frac{p_b}{p^+} \right)^{1/\gamma}, \quad u_{n,b} = u_n^+ + \frac{2}{\gamma - 1} (a^+ - a_b), \quad \mathbf{v}_{t,b} = \mathbf{v}_t^+. \quad (31)$$

The two branches are selected by the local normal Mach number, so a shock crossing the exit plane during the transient is handled without special cases.

The legacy solver had no back pressure. Its outflow condition always extrapolated, so the `p_back_ratio` input never reached the flow. Every operating point with a shock in the nozzle — that is, most of the interesting ones — was therefore unreachable.

7 Time integration

Equation (19) is marched in pseudo-time to steady state with an element-local step,

$$\Delta t_e = \frac{\text{CFL}}{2p+1} \frac{2A_e}{\sum_{f \in \partial\Omega_e} s_f \ell_f}, \quad (32)$$

where A_e is the element area, ℓ_f the length of face f and s_f the maximum signal speed there. The $1/(2p+1)$ factor is the standard order-dependent restriction for explicit DG.

Two schemes are available. `'rk4'` is the classical four-stage method, with the local step frozen at the first stage (the intermediate stages are not meant to be time-accurate). `'ssprk3'` is the three-stage strong-stability-preserving method,

$$\begin{aligned} \mathbf{U}^{(1)} &= \mathbf{U}^n + \Delta t \mathcal{L}(\mathbf{U}^n), \\ \mathbf{U}^{(2)} &= \frac{3}{4}\mathbf{U}^n + \frac{1}{4}\left(\mathbf{U}^{(1)} + \Delta t \mathcal{L}(\mathbf{U}^{(1)})\right), \quad \mathcal{L} = -M^{-1}\mathbf{R}. \\ \mathbf{U}^{n+1} &= \frac{1}{3}\mathbf{U}^n + \frac{2}{3}\left(\mathbf{U}^{(2)} + \Delta t \mathcal{L}(\mathbf{U}^{(2)})\right), \end{aligned} \quad (33)$$

Each SSP stage is a convex combination of forward-Euler updates, which is exactly the structure the positivity limiter assumes — so `'ssprk3'` plus `'positivity'` carries a genuine positivity guarantee that `'rk4'` does not.

Convergence measure. The march stops when

$$\frac{\|\dot{\mathbf{U}}\|_{\text{rms}}}{\rho_t a_t / L} < \text{tolerance}, \quad \|\dot{\mathbf{U}}\|_{\text{rms}} = \sqrt{\frac{1}{4N} \sum (M^{-1}\mathbf{R})^2}. \quad (34)$$

The choice of denominator matters more than it looks. Testing the *unnormalised* residual against a fixed threshold — what the legacy solver did — makes “converged” mean different things on different meshes and at different operating points. But testing it *relative to the first iteration* is no better: it demands a tighter absolute residual the better the initial guess is, so improving the initial condition makes the solver appear slower and two runs started differently cannot be compared at all. A fixed physical scale, $\rho_t a_t / L$, is independent of both the mesh and the starting field. The legacy solver additionally tested the *fourth* Runge–Kutta stage’s rate, which is not the residual at any state it holds.

8 Limiting

At $p = 0$ the solution is constant per element and cannot oscillate. At $p \geq 1$ it can, and near a shock it does: the polynomial overshoots, pressure or density goes negative at a quadrature point, and the Roe flux fails.

Both limiters act by shrinking the deviation from the cell average,

$$\mathbf{U}_{e,i}^{\text{lim}} = \bar{\mathbf{U}}_e + \theta_e (\mathbf{U}_{e,i} - \bar{\mathbf{U}}_e), \quad \theta_e \in [0, 1], \quad (35)$$

which leaves the cell average untouched and is therefore *conservative*.

Positivity (Zhang–Shu). θ_e is the largest value keeping ρ and p above small floors at every point the scheme evaluates — the volume quadrature points and the element’s own face quadrature points. The density constraint is linear in θ and solved in closed form; the pressure constraint is quadratic and bracketed by bisection. On a smooth solution $\theta_e = 1$ everywhere and the limiter is exactly inactive, so full accuracy is retained.

Barth–Jespersen. Additionally bounds the solution by the neighbouring cell averages, damping the oscillations themselves rather than only their consequences, at the cost of clipping genuine extrema.

When a limiter cannot help. If the *cell average* itself becomes non-physical, no scaling can repair it — at $\theta = 0$ the solution *is* that average. The code then floors the average minimally, counts the repair, and reports it; a run needing repairs should be rerun with a smaller `cfl`. Separately, a limiter that switches on and off between iterations parks the residual at a fixed level instead of converging. The solver detects that limit cycle, stops, and says so.

9 Quasi-one-dimensional reference solution

Quasi-1D theory serves two purposes: it is an exact reference to compare against, and it is the initial condition. As a starting field it saves a modest number of iterations (about 15% at $p = 0$, 3% at $p = 1$ on the reference case); its real value is robustness, since at $p = 2$ without p -continuation a uniform start diverges where the quasi-1D start converges.

The area–Mach relation is

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}, \quad (36)$$

monotone on each of the subsonic and supersonic branches and therefore invertible by bracketed bisection. The isentropic relations give

$$\frac{p}{p_t} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma}{\gamma - 1}}, \quad \frac{\rho}{\rho_t} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{1}{\gamma - 1}}, \quad \frac{T}{T_t} = \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-1}. \quad (37)$$

Across a normal shock,

$$M_2^2 = \frac{1 + \frac{\gamma - 1}{2} M_1^2}{\gamma M_1^2 - \frac{\gamma - 1}{2}}, \quad \frac{p_2}{p_1} = \frac{2\gamma M_1^2 - (\gamma - 1)}{\gamma + 1}, \quad \frac{A_2^*}{A_1^*} = \frac{p_{t1}}{p_{t2}}. \quad (38)$$

9.1 Operating regimes

Three critical back-pressure ratios divide the operating map:

Ratio	Condition	Regime below it
$(p_b/p_t)_1$	$M = 1$ at the throat, subsonic exit	choked, shock in the diverging section
$(p_b/p_t)_2$	normal shock at the exit plane	shock-free in the nozzle, over-expanded
$(p_b/p_t)_3$	shock-free supersonic (design)	under-expanded

Table 2: Critical pressure ratios. Above $(p_b/p_t)_1$ the nozzle is not choked and the mass flow still depends on back pressure.

For the reference geometry ($AR = 2.50189$, $\gamma = 1.4$) these are 0.9609, 0.4345 and 0.06390, with a design exit Mach number of 2.4436.

When a shock stands in the diverging section its position is found by bisection: p_{exit} falls monotonically as the shock moves downstream, so there is exactly one position at which it equals p_b .

10 Performance metrics

10.1 Thrust

For steady inviscid flow the axial momentum balance over the nozzle interior closes exactly, giving *two* expressions for the same force:

$$F = \underbrace{\int_{\text{exit}} (\rho u^2 + p) \, dA - \int_{\text{inlet}} (\rho u^2 + p) \, dA}_{\text{momentum form}} = - \underbrace{\oint_{\text{wall}} p n_x \, ds}_{\text{wall form}}. \quad (39)$$

They are identical in exact arithmetic but *discretely* distinct, so the gap between them is a free measure of discretisation error. Both are computed and reported, together with

$$c_F = \frac{F}{p_t A_{\text{throat}}}, \quad A_{\text{throat}} = 2y_{\text{th}}. \quad (40)$$

The legacy normalisation was wrong for any modified geometry. It divided by a hard-coded throat height $h = 0.13989434$ — the throat of the original contour — so every thrust coefficient computed at a different `area_ratio` or `throat_x` used the wrong reference length. It also integrated the wall pressure over every face tagged `-2`, which in that mesh included the symmetry axis.

10.2 Entropy error

For shock-free flow the exact solution is isentropic, so

$$\varepsilon_s = \sqrt{\frac{1}{A} \int_{\Omega} \left(\frac{s}{s_t} - 1 \right)^2 \, d\Omega}, \quad s = \frac{p}{\rho^\gamma}, \quad s_t = p_t^{1-\gamma} (RT_t)^\gamma, \quad (41)$$

is a genuine error measure that converges at the scheme’s design rate. Across a shock, entropy rises physically, and the number then reflects the flow rather than the discretisation.

11 Verification

11.1 Exactness checks

The following hold to machine precision and are asserted in the test suite:

- **Geometric conservation.** For a uniform state with the exact normal flux imposed on every edge, the total residual is $\mathcal{O}(10^{-17})$ — the volume gradient term and the face term cancel identically. This is the strongest available check on the metric terms.
- **Discrete divergence theorem.** $\sum_f \mathbf{n}_f \ell_f = \mathbf{0}$ per element to 10^{-12} , for triangles and quadrilaterals, $Q = 1$ and $Q = 2$.
- **Flux consistency.** $\hat{F}(\mathbf{U}, \mathbf{U}, \mathbf{n}) = \mathbf{F} \cdot \mathbf{n}$, and $\hat{F}(\mathbf{U}^+, \mathbf{U}^-, \mathbf{n}) = -\hat{F}(\mathbf{U}^-, \mathbf{U}^+, -\mathbf{n})$.
- **Inflow boundary condition.** A uniform isentropic state is reproduced exactly, to 10^{-15} , for $M \in [0.05, 0.9]$.
- **Backend agreement.** The Numba and NumPy residuals agree to a relative 4×10^{-15} .
- **Order continuation.** Re-projection $p \rightarrow p + 1 \rightarrow p$ returns the original coefficients to 10^{-15} , and preserves cell averages exactly.
- **Limiter conservation.** Cell averages are bit-identical before and after limiting whenever the average is admissible.

11.2 Order of accuracy

Table 3 reports the entropy error, Eq. (41), on uniformly refined meshes at the shock-free design point, with curved ($Q = 2$) elements.

Contour	p	DOF	ε_s	observed rate
analytic	1	420	1.904×10^{-3}	—
	1	1680	4.996×10^{-4}	1.93
	1	6720	1.280×10^{-4}	1.96
	2	840	1.871×10^{-4}	—
	2	3360	2.872×10^{-5}	2.70
smooth	1	420	1.506×10^{-3}	—
	1	1680	3.835×10^{-4}	1.97
	1	6720	9.941×10^{-5}	1.95
	2	840	1.587×10^{-4}	—
	2	3360	2.635×10^{-5}	2.59
bell	1	420	3.835×10^{-3}	—
	1	1680	7.780×10^{-4}	2.30
	1	6720	7.731×10^{-4}	0.01

Table 3: Observed orders of accuracy, $Q = 2$, $p_b/p_t = 0.15$. The expected rates are $p + 1$.

The C^1 contours achieve their design rates. The ‘bell’ contour *stalls*: its 21.3° throat corner launches a Prandtl–Meyer expansion fan, a genuine singularity in the exact solution, and no amount of mesh refinement recovers the asymptotic rate. This is physics, not a solver defect — but it means *a convergence study must use ‘smooth’ or ‘analytic’*.

The thrust functional converges cleanly in all cases. The gap between the two forms of Eq. (39) falls as $9.7 \times 10^{-3} \rightarrow 2.7 \times 10^{-3} \rightarrow 6.6 \times 10^{-4}$ for $p = 1$, $Q = 1$ under refinement (rate ≈ 1.9), and reaches 6.7×10^{-5} at $p = 2$, $Q = 2$.

11.3 Effect of curved elements

At fixed element count, raising the geometry order from $Q = 1$ to $Q = 2$ reduces the error in the discrete domain area from 1.10×10^{-3} to 8.76×10^{-6} — a factor of 125 — because the curved wall is represented quadratically rather than by chords. At $p = 2$ the entropy error improves correspondingly, from 1.30×10^{-3} to 2.91×10^{-4} . With $Q = 1$ the geometry error dominates and caps the observed rate near 1.6 regardless of p .

12 Design sensitivities

For a scalar functional J and design variables $\mathbf{a}$, the converged state satisfies $\mathbf{R}(\mathbf{U}, \mathbf{a}) = \mathbf{0}$. Differentiating implicitly,

$$\frac{dJ}{d\mathbf{a}} = \frac{\partial J}{\partial \mathbf{a}} - \frac{\partial J}{\partial \mathbf{U}} \left(\frac{\partial \mathbf{R}}{\partial \mathbf{U}} \right)^{-1} \frac{\partial \mathbf{R}}{\partial \mathbf{a}} = \frac{\partial J}{\partial \mathbf{a}} - \boldsymbol{\lambda}^T \frac{\partial \mathbf{R}}{\partial \mathbf{a}}, \quad (42)$$

where the adjoint variable solves

$$\left(\frac{\partial \mathbf{R}}{\partial \mathbf{U}} \right)^T \boldsymbol{\lambda} = \left(\frac{\partial J}{\partial \mathbf{U}} \right)^T. \quad (43)$$

The cost is one adjoint solve *regardless of the number of design variables* — where finite differences would need two flow solves each.

Equation (43) is solved by pseudo-time march,

$$\frac{d\boldsymbol{\lambda}}{d\tau} = -M^{-1} \left((\partial \mathbf{R} / \partial \mathbf{U})^T \boldsymbol{\lambda} - (\partial J / \partial \mathbf{U})^T \right), \quad (44)$$

mirroring the forward solver. Because M is symmetric positive definite, $M^{-1}(\partial \mathbf{R} / \partial \mathbf{U})^T$ is similar to $M^{-1} \partial \mathbf{R} / \partial \mathbf{U}$ and has the same spectrum, so the forward solver’s time step and CFL limit carry over unchanged and no preconditioner is needed. Both Jacobian actions come from automatic differentiation of the residual; no matrix is ever assembled.

Mesh consistency. The gradient is taken on a *fixed logical grid*. This matters: with throat-following node clustering, rebuilding the mesh at a perturbed x_{th} also redistributes the nodes, so a finite difference would measure node motion as well as the design change. Freezing the logical grid removes that inconsistency.

Verification. Against central differences with a relative step of 10^{-4} , at $p = 1$, ‘smooth’, $p_b/p_t = 0.15$:

Parameter	adjoint dF/da	finite difference	relative error
area_ratio	9.772082×10^{-3}	9.772082×10^{-3}	2.1×10^{-8}
throat_x	-1.490300×10^{-2}	-1.490297×10^{-2}	2.0×10^{-6}
theta_exit	7.602688×10^{-3}	7.602689×10^{-3}	9.2×10^{-8}
inlet_half_height	-1.524949	-1.524949	7.7×10^{-8}
length	1.353855×10^{-2}	1.353855×10^{-2}	5.6×10^{-8}

Table 4: Adjoint gradients of thrust, verified against finite differences.

Validity. The gradient is of the *discrete* problem, which is what gradient-based optimisation needs. Two caveats: the geometry must be smooth in $\mathbf{a}$ (all contour families here are), and the flow should be shock-free with the limiter inactive — a limiter switching on and off introduces kinks, and a shock’s position is only piecewise differentiable in the design variables.

13 Implementation

13.1 Assembly as a pure gather

Every (element, local face) pair maps to exactly one global edge. Face integrals are therefore collected by *indexing* rather than by scatter-add. Three consequences: the loops vectorise, they parallelise without atomics, and the identical code runs under NumPy, Numba and JAX.

13.2 Backends

Backend	Role
numba	Compiled nopython kernels with <code>prange</code> over edges and elements. The default and the fastest.
numpy	Vectorised reference. No compile step; the correctness baseline for the other two.
jax	The same vectorised source under <code>jax.numpy</code> , jit-compiled with the iteration loop fused into a <code>lax.fori_loop</code> . Differentiable, hence the adjoint.

Table 5: Execution backends.

13.3 Cost and robustness measures, none of which change the answer

- **Quasi-1D initial condition** (on by default) — the shock is roughly in place and the nozzle already choked. The iteration saving is small (about 15% at $p = 0$, 3% at $p = 1$); what it buys is convergence at $p = 2$, where a uniform start diverges outright.
- **p -continuation** (off by default) — solve at $p = 0$, re-project upward one order at a time. Re-projection is exact, because the coarse space sits inside the fine one, so it cannot change the answer. It is *not* a speed-up: measured, it costs 2–28% more wall time than a direct solve, the quasi-1D start having already removed the transient the $p = 0$ stage exists to remove. Its value is as a fallback — it converges $p = 2$ from a uniform initial condition, which a direct solve does not.
- **Warm-started sweeps** — the mesh topology is invariant under design changes, so the previous point’s field is always a valid start.
- **Limiter fast path** — on a smooth solution no element violates positivity, so the pressure bisection is skipped entirely. This is the single largest saving in the code: the limiter runs once per Runge–Kutta stage, and in naive vectorised form it cost about twenty times a residual evaluation and dominated the solve. Moving it into a kernel with a zero-allocation fast path took one RK4 step from 3.92 ms to 0.34 ms.

On measuring these honestly. The first two were documented as large speed-ups until they were measured. The iteration counts that appeared to support that were an artefact of two defects, both since fixed: convergence tested *relative to the first residual*, which tightens the target as the initial guess improves and so penalises a good starting field; and p -continuation reporting only its final stage’s iteration count, which hid the cost of the low-order stages that preceded it. A performance claim that has not survived an attempt to measure it is not a performance claim.

14 Summary of corrections to the legacy solver

Issue	Consequence
Diverging-contour rescale used constants evaluated at the wrong parameter values	The default 'smooth' contour reached an exit half-height of 4.34 m instead of 0.350 m, and went <i>negative</i> just past the throat — inverted elements.
Outflow always extrapolated	<code>p_back_ratio</code> was dead; no shocked operating point was reachable.
Inflow quadratic root selection	Returned a negative Mach number where the leading coefficient changed sign.
Hard-coded throat height in the thrust normalisation	Wrong c_F for any geometry but the original.
Wall integral included the symmetry axis	Harmless for a horizontal axis, wrong in general.
<code>extrapolate</code> read the new order as the old one	Order continuation indexed past the end of its input; the mesh also changed between calls, so element counts disagreed.
<code>postprocess()</code> took no arguments but was called with two; loaded <code>.mat</code> files that were never written	Post-processing could not run at all.
Dunavant degree-3 rule (negative weight)	Risked an indefinite mass matrix and a negative sum-of-squares integral.
Unnormalised convergence test on the fourth RK stage; no iteration cap	“Converged” meant different things on different meshes; a diverging run never stopped.
Negative Roe sound speed raised an error	A recoverable transient aborted the whole run.

Table 6: Defects found in the MATLAB implementation and fixed here.

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